Highlights

Enhancing Multi-Agent Large Language Models for Stock Trading through Quantitative Alpha Integration[⋆]

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- Quantitative alpha and Vietnamese sentiment extend a multi-agent trading framework.
- An 85-factor registry adapts OHLCV signals to single-stock decision context.
- Shared-report comparisons evaluate factor, sentiment and past-donor variants.
- Balanced-accuracy gains coincide with fewer false LONG forecasts.
- Cost-aware accounts distinguish directional scores from economic outcomes.

Enhancing Multi-Agent Large Language Models for Stock Trading through Quantitative Alpha Integration

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Abstract

Quantitative signals and local-language news offer complementary context for multi-agent LLM trading, but their incremental value requires matched evaluation. We study quantitative alpha integration in a five-role framework for Vietnamese equities, combining an 85-candidate OHLCV registry, Vietnamese sentiment and two chart-reading agents. Selected factors and sentiment remain separate inputs to a report-based Decision Agent. A shared-report protocol compares Full and Base on 102 historical market units across nine equities, twelve dates and six blocks, with 25 UP and 77 DOWN net-return labels. Full balanced accuracy is 52.96%, versus 49.71% for Base and 55.61% for a price-only logistic comparator. The Full–Base difference is 3.25 percentage points, with a descriptive block interval of $[-1.18, 8.13]$ points. Both variants capture seven UP outcomes; Full makes fewer false LONG forecasts. Mean separate-account net returns are -0.797% and -0.750%, respectively. Three-execution component/placebo panels and train-fit numeric controls provide additional, separately scoped comparisons. The reduced historical scope is exploratory, native factor construction is not fully row-causal, and financial-news validity and availability remain unverified. The study contributes a specified integration and comparative evaluation, while its results do not establish causal component gains or deployable profitability.

Keywords: Multi-agent LLM, Knowledge fusion, Knowledge integration, Financial decision support, Alpha factor selection

1. Introduction

Large language models (LLMs) have recently been used as reasoning engines in financial analysis. Their value for knowledge-based decision support depends on how quantitative, textual and visual evidence is represented and assessed, rather than on distributing tasks among agents alone. Frameworks such as QuantAgent [1] provide a starting point for chart-based LLM decision support. Adapting such a framework to Vietnamese equities requires explicit treatment of Vietnamese-language news, single-asset factor inputs and the distinction between a directional forecast and a feasible cash-equity trade.

Vietnam motivates this implementation in three ways. First, the bar-count forecast horizon must be distinguished from entry, exit and settlement assumptions. Second, Vietnamese news requires

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a language-appropriate scorer, whose financial-domain validity still needs independent assessment. Third, OHLCV-only factor construction cannot retain the peer-relative meaning of every cross-sectional expression. These are practical adaptation questions, not evidence that the resulting system has solved emerging-market prediction.

We examine a concrete multi-agent implementation for Vietnamese equities. It combines deterministic technical-indicator reports, a native selector over 85 OHLCV factor candidates, Vietnamese sentiment scores, two chart-reading roles and a final binary Decision Agent. Indicator, Pattern, Trend and Decision roles and associated utilities reuse or modify QuantAgent components [1]. The integration is an implementation and domain-adaptation exercise: condensed reports enter a prompt with heuristic evidence-priority instructions, rather than a learned knowledge-fusion representation.

Our question is how this implementation compares with class-aware and numerical controls on the same recorded historical market units, and what its archived executions can support about variability, components and economics. The primary comparison is Full versus Base. Base removes both selected-factor and sentiment sections, so this is a bundle contrast, not an isolated alpha-information treatment. A separate factorial panel includes factor-only, sentiment-only and matched-structure past-donor variants. Different prompt families and small repeated panels remain explicit rather than being pooled into a larger apparent experiment.

The completed release contains 102 main market units spanning six historical blocks and twelve selected dates, with 25 UP and 77 DOWN net labels. Full has pooled balanced accuracy (BA) of 52.96%, Base 49.71%, and a price-only logistic comparator 55.61%. CASH has higher ordinary accuracy than Full but BA of 50%. Mean reference account returns are negative for Full and Base. These observations motivate an assessment of the implementation and its limits, rather than a claim of profitable trading or general predictive superiority.

The evaluation also exposes unresolved methodological defects. Native factor operations are isolated from prices after the outer decision cutoff in synthetic probes, but are not causal row by row within their trailing history. Native ranking and orientation diagnostics are in-sample. News availability and the sentiment classifier's financial-domain validity are not independently authenticated. We preserve these facts when reporting the frozen outputs. Separate train-fit numeric controls use causal adapters and cannot retrospectively repair the native LLM study.

The main contributions are as follows:

- A Vietnamese-equity adaptation that integrates an OHLCV factor selector and local-language sentiment with Indicator, Pattern, Trend and Decision roles. It extends reused chart-agent components through explicit quantitative and textual inputs rather than claiming a new learned fusion architecture.

- A specified single-asset signal-construction method: 80 retained Formulaic Alpha adaptations and five custom handlers, top-five diagnostic ranking, and a separately represented time-decayed sentiment report. Its operator semantics and construction limits make the actual inputs distinguishable from idealised source formulas.

- A shared-report evaluation design combining common-support baselines, repeated upstream and Decision executions, and five variants for factors, sentiment and a past donor. The comparison identifies which inputs differ without treating a bundled contrast as a causal alpha effect.

- An empirical analysis linking class-aware prediction to transaction-cost-aware long-only accounts. On the main support, the Full–Base balanced-accuracy difference accompanies fewer

false LONG forecasts and unchanged UP recall, while mean net returns remain negative. This distinguishes a classification change from demonstrated economic value.

Historical snapshots, actual requests/responses and metrics are retained with hashes for offline replay. The evidence is historical and exploratory: the reduced scope was amended after some inference, all outcome dates preceded the freeze, and the study does not establish causal component value or prospective deployment performance.

The remainder of this paper is organised as follows. Section 2 reviews related work; Section 3 formalises the task and signals; Section 4 describes each agent; Section 5 specifies the evaluation; Section 6 reports results; Section 7 discusses limitations and future work; and Section 8 concludes.

2. Related works

2.1. LLM Agents in Financial Analysis

Financial language models and adaptation frameworks include BloombergGPT and FinGPT [2, 3]. TradingAgents organizes multiple financial analysis and decision roles [4]. QuantAgent's version-specific architecture already includes Indicator, Pattern, Trend and Decision agents, shared state, chart rendering and trendline utilities [1]. Our system reuses or modifies those elements; adding an alpha/sentiment path does not make the underlying chart-agent architecture novel.

2.2. Technical Analysis and Pattern Recognition

ChartLlama addresses multimodal chart understanding [5], while the cited Tsantekidis study concerns limit-order-book prediction, not candlestick-image recognition [6]. These tasks should not be treated as interchangeable validation of our chart reader.

2.3. Quantitative Alpha Factor Research

The source catalogue of 101 Formulaic Alphas describes expressions with cross-sectional and other requirements [7]. Our 80 retained expressions are heuristic single-asset adaptations, with operators and deviations documented individually; five further formulas are custom to the implementation. Searching many candidates and choosing their orientation on the same history creates selection bias [8]. Nested selection and assessment [9], the Deflated Sharpe Ratio [10], and White's Reality Check [11] address related evaluation concerns. This study does not claim to have applied all those procedures: its auxiliary numeric study uses train-fit controls and a declared Holm family, while the native selector retains its audited defects.

2.4. Sentiment Analysis in Finance and Vietnamese NLP

ViSoBERT was developed for Vietnamese social-media language understanding [12]. We use a distinct public 5CD-AI sentiment classifier derived from that backbone [13]; its identity does not establish accuracy on Vietnamese financial news. General financial sentiment work such as FinBERT [14] does not substitute for validating this classifier and corpus. Accordingly, we report the implemented scoring recipe, availability and provenance rather than assuming financial-domain competence.

2.5. Trading Systems for Emerging Markets

The combination of local-language information, purposive instrument coverage and cash-equity execution makes a Vietnamese application different from a generic English-language trading demonstration. A local implementation nevertheless needs the same tests of class balance, information availability, selection bias and costs. Neither using Vietnamese text nor adding a factor report establishes that an imported architecture generalizes to a wider emerging-market universe. Our contribution is the documented adaptation and evaluation on the declared nine equities, not a claim that existing systems lack vision, factor research or multi-agent coordination.

2.6. Evaluation Standards and Agentic Benchmarks

Shared-report comparisons control the upstream report snapshot within a pair, while repeated executions measure some hosted variability. They do not create independent market dates or remove information-set and exposure differences. Accuracy, balanced accuracy, both-class support and economic outcomes consequently answer different evaluation questions.

Exact hosted-model replay poses an additional limitation: the text model's published description [15] does not identify the immutable weights served for an archived request. We retain observed aliases and fingerprints without treating them as weight attestations.

3. Problem Formulation

Algorithm 1: Walk-Forward Evaluation

Input: OHLCV series $\mathbf{X}$, alpha registry $\mathcal{A}$, sentiment context $\mathcal{D}$, chart window W, lookahead L, frozen eligible test schedule $\mathcal{T}$ and variant masks

Output: Directional predictions $\hat{y}_{e,L}$, class-aware paired metrics and separate-account returns

Initialize results list $\mathcal{R} \leftarrow \emptyset$;
foreach *test point e in chronological order in $\mathcal{T}$* **do**
  Extract chart window $\mathbf{X}_e = \mathbf{X}_{[e-W,e)}$ and alpha history ending at $d = e - 1$;
  Compute deterministic technical reports and native candidate diagnostics;
  Form eligible candidates $\mathcal{V}$ and rank by;

$$S_k = 0.40\mathcal{N}_{\mathcal{V}}(|IC_k|) + 0.35\mathcal{N}_{\mathcal{V}}(Acc_k) + 0.25\mathcal{N}_{\mathcal{V}}(Sharpe_k)$$

  Select top-five $\mathcal{A}^*$; obtain separate sentiment aggregate/count context;
  Generate the two charts from the same $\mathbf{X}_e$ and retain their exact inputs;
  Run or retrieve the recorded Indicator, Pattern and Trend reports shared within the pair;
  foreach *registered variant* **do**
    Compose Decision input from shared reports and enabled factor/sentiment sections;
    Retain its actual LONG/SHORT request, output and execution provenance;
  Join predictions to common net-return labels; update the separate long-only reference ledgers;
  Store outcomes in $\mathcal{R}$;
Compute pooled/macro classification, paired contrasts and account outcomes on their declared support;
return $\mathcal{R}$

3.1. Market Model and Data

Let $\mathcal{S} = \{s_1, s_2, \ldots, s_N\}$ denote the declared universe of Vietnamese equities. For each asset $s \in \mathcal{S}$, the market is observed as a discrete-time OHLCV series:

$$\mathbf{x}_u = \Big(O_u,\, H_u,\, L_u,\, C_u,\, V_u\Big) \;\in\; \mathbb{R}_+^5, \quad u = 0, 1, \ldots, T-1, \tag{1}$$

Here O_u, H_u, L_u, C_u, V_u denote Open, High, Low, Close and vendor volume at zero-based bar position u. Missing volume may trigger an intrabar-range proxy in the native feature construction; this is an implementation substitution, not measured liquidity. The historical volume-availability branch is evaluated over the supplied frame and is included in the causal limitation below.

We use zero-based row positions and an exclusive window-end boundary. For a test point with boundary e, the analysis window of size W is exactly:

$$\mathbf{X}_e = \Big(\mathbf{x}_{e-W},\, \mathbf{x}_{e-W+1},\, \ldots,\, \mathbf{x}_{e-1}\Big) \;\in\; \mathbb{R}^{W\times 5}. \qquad (\text{equivalently, } \mathbf{X}_e = \mathbf{X}_{[e-W,e)}) \tag{2}$$

Thus e is not itself a visible bar: the final visible decision bar is $d = e - 1$. The next bar e is reserved for the earliest simulated entry, and the target is $h = e - 1 + L$. The time-index contract is

$$\boxed{\text{window} = [e-W, e), \qquad d = e-1, \qquad \text{entry} = e, \qquad h = e-1+L.} \tag{3}$$

The mapping is positional rather than calendar arithmetic. A weekend, holiday or vendor gap does not create a synthetic bar. The first complete window is $e = W$ and requires $T \geq W + L$ for a complete target. For $L = 1$, the target is the entry bar; for $L = 3$, it is two positions after entry. Incomplete targets are rejected rather than shortened. Charts use $W = 45$; the alpha path separately receives up to 600 observed bars ending at the same decision cutoff. We denote that prefix by $\mathbf{X}_e^{\text{hist}}$.

3.2. Directional Prediction Task

The core task predicts whether a candidate LONG cycle has positive net return after both transaction legs. With $d = e - 1$ and $h = e - 1 + L$, define

$$r_{e,L}^{\text{net}} = \frac{C_h}{O_e}\frac{(1-c_{\text{slip}})(1-c_{\text{fee}})}{(1+c_{\text{slip}})(1+c_{\text{fee}})} - 1, \qquad y_{e,L} = \begin{cases} +1, & r_{e,L}^{\text{net}} > 0, \\ -1, & r_{e,L}^{\text{net}} \leq 0. \end{cases} \tag{4}$$

Here O_e is the next-bar entry Open and C_h the target Close. The costs are $c_{\text{fee}} = 0.0025$ and $c_{\text{slip}} = 0.001$ on each leg. Zero net return belongs to DOWN. The reported study uses daily bars with $L = 3$; the three-bar horizon includes the entry bar. The mathematical coding $+1/-1$ denotes the same UP/DOWN target as the exported binary labels; no prediction or target is recoded to change an outcome.

A trading system $\mathcal{F}$ maps the analysis window and its associated historical context to a binary decision. Suppressing the extra context arguments for compact notation,

$$\hat{y}_{e,L} \;=\; \mathcal{F}\Big(\mathbf{X}_e\Big) \;\in\; \{-1,\, +1\}, \tag{5}$$

where $+1$ corresponds to `LONG` and -1 to `SHORT`. In the long-only reference ledger, SHORT maps to CASH, not a short sale; a correct DOWN forecast earns no bearish trading return. Account return therefore differs from the candidate-LONG target. Gross returns are a separate diagnostic on the same net-elicited predictions, not a replacement target. Decision Close differs from entry Open at 65/102 units, and seven gross/net labels differ. A legacy field called `actual_prev_close` has unresolved price semantics; canonical evaluation uses explicit decision/entry/exit fields.

3.3. Alpha Factor Formalisation and Adaptation

An alpha factor $\alpha^{(k)}$ maps an engineered feature frame from the historical prefix to a scalar signal:

$$\alpha^{(k)} : \mathbb{R}^{T_{\text{hist}} \times d} \longrightarrow \mathbb{R}, \tag{6}$$

where d in this mapping is the feature dimension and T_{hist} is the computation history length. The native registry contains 80 retained WorldQuant adaptations and five custom formulas, for 85 candidates. These single-asset translations are engineering heuristics rather than mathematically equivalent cross-sectional alphas [7]. Among the 21 excluded source expressions, 17 use industry neutralisation and one market capitalisation; three price/volume omissions lack an authenticated selection rationale. They are not all characterised as impossible cross-sectional translations.

Operator Localisation. A cross-sectional rank is localised to a 20-bar own-history rank unless an expression specifies its own time-series rank/lookback. With preceding observations $\mathcal{H}_{u,w}$ and $n_{u,w} = |\mathcal{H}_{u,w}| + 1$, the implementation computes

$$\text{ts_rank}(x_u, w) \;=\; \frac{1}{\max(n_{u,w} - 1, 1)} \sum_{x_v \in \mathcal{H}_{u,w}} \mathbf{1}[x_u > x_v]. \tag{7}$$

Ties are not counted as strictly lower values. The operator permits partial windows subject to its minimum observations; other expression-specific windows can be much longer than 20 bars. ADV is mean share volume, VWAP is a daily typical-price proxy, and percentage changes, differences and lags retain implementation-specific fills/guards. These substitutions remove peer-relative information and can alter scale, persistence and economic meaning.

Consistent Normalisation. The zscore–tanh path normalises a raw factor as

$$\tilde{\alpha}^{(k)}_{u,e} \;=\; \tanh\!\left(\frac{\alpha^{(k)}_u - \mu_{k,e}}{\sigma_{k,e} + \varepsilon}\right), \tag{8}$$

where $\mu_{k,e}$ and $\sigma_{k,e}$ are obtained from finite values in the supplied historical prefix and $\varepsilon > 0$ guards the denominator. Min–max and rank-based alternatives also occur. This equation describes one native transform, not a training-only fit guaranteed causal at every row. Prefix-dependent location/scale can change earlier values when the supplied history changes. Values are interpreted as directional scores, not calibrated probabilities.

3.4. Walk-Forward Alpha Selection

Let $\mathcal{A} = \{\alpha^{(1)}, \ldots, \alpha^{(M)}\}$ with $M = 85$. The native candidate diagnostics estimate association, orientation and ranking on the same trailing observations. They are in-sample selection statistics, not independent discoveries. For a signal row u, write $r^{\text{net},(L)}_u = r^{\text{net}}_{u+1,L}$ and $y^{(L)}_u = y_{u+1,L}$, so the diagnostic uses entry at the next bar rather than the signal bar. For each candidate's finite evaluation sample $\mathcal{T}_{\text{eval}}$, the Information Coefficient is

Information Coefficient (IC).

$$\text{IC}_k \;=\; \text{SpearmanCorr}\Big(\tilde{\alpha}^{(k)}_{\mathcal{T}_{\text{eval}}},\; r^{\text{net},(L)}_{\mathcal{T}_{\text{eval}}}\Big). \tag{9}$$

The IC uses the pre-orientation signal. A negative IC reverses the signal for directional diagnostics, the signed-return proxy and current report. Write $\tilde{\alpha}^{(k),\text{or}}_u = o_k \tilde{\alpha}^{(k)}_u$, where $o_k = -1$ for negative IC and $o_k = 1$ otherwise. Sign choice and ranking reuse the same observations; outer-cutoff truncation is not a substitute for an independent assessment.

Directional Accuracy. For the strong-signal sample $\mathcal{T}^* = \{u : |\tilde{\alpha}_u^{(k),\mathrm{or}}| > 0.05\}$, the directional diagnostic is

$$\mathrm{Acc}_k \;=\; \frac{1}{|\mathcal{T}^*|} \sum_{u \in \mathcal{T}^*} \mathbf{1}\Big[\mathrm{sign}\Big(\tilde{\alpha}_u^{(k),\mathrm{or}}\Big) = y_u^{(L)}\Big], \tag{10}$$

The native guard returns 0.5 when fewer than ten strong signals are available. With fewer than twenty finite paired observations, it returns default diagnostics and zero diagnostic support; the selector's finite-support policy must be distinguished from a causal warm-up trim.

Long-Only Win Rate. The bullish-signal diagnostic is

$$\mathrm{LongAcc}_k \;=\; \frac{\sum_u \mathbf{1}\Big[\tilde{\alpha}_u^{(k),\mathrm{or}} > 0.05 \;\wedge\; r_u^{\mathrm{net},(L)} > 0\Big]}{\sum_u \mathbf{1}\Big[\tilde{\alpha}_u^{(k),\mathrm{or}} > 0.05\Big]}. \tag{11}$$

The implementation uses a default of 0.5 for this diagnostic when fewer than five bullish signals are available. It is not a realised trading return or an out-of-sample precision estimate.

Horizon Sharpe diagnostic. Define $z_u^{(k)} = \mathrm{sign}(\tilde{\alpha}_u^{(k),\mathrm{or}}) r_u^{\mathrm{net},(L)}$ on the finite trailing sample. The native ranking proxy is

$$\mathrm{Sharpe}_k \;=\; \sqrt{\frac{252}{L}} \cdot \frac{\mathbb{E}_{u \in \mathcal{T}_{\mathrm{eval}}}\Big[z_u^{(k)}\Big]}{\mathrm{Std}_{u \in \mathcal{T}_{\mathrm{eval}}}\Big[z_u^{(k)}\Big] + \varepsilon}. \tag{12}$$

Its implementation uses population SD and $\varepsilon = 10^{-9}$. The $\sqrt{252/L}$ factor is part of this historical ranking proxy; it does not annualise the irregular account schedule or represent an executable short strategy. Other diagnostics such as Sortino, drawdown, hit rate and long-only accuracy do not enter the composite score.

Eligible candidates $\mathcal{V} \subseteq \mathcal{A}$ are scaled candidate-wise for $q \in \{|IC|, Acc, Sharpe\}$:

$$\mathcal{N}_{\mathcal{V}}(q_k) = \begin{cases} \dfrac{q_k - q_{\min}}{q_{\max} - q_{\min}}, & q_{\max} - q_{\min} \geq \varepsilon_{\mathrm{rank}}, \\ \dfrac{1}{2}, & \text{otherwise}, \end{cases} \tag{13}$$

Here $q_{\min}$ and $q_{\max}$ are extrema over $\mathcal{V}$ and $\varepsilon_{\mathrm{rank}} = 10^{-9}$. A constant range maps to 0.5. The default ranking is

$$S_k \;=\; 0.40\,\mathcal{N}_{\mathcal{V}}(|\mathrm{IC}_k|) \;+\; 0.35\,\mathcal{N}_{\mathcal{V}}(\mathrm{Acc}_k) \;+\; 0.25\,\mathcal{N}_{\mathcal{V}}(\mathrm{Sharpe}_k). \tag{14}$$

The displayed diagnostic rounding precedes sorting. Up to $K = \min(5, |\mathcal{V}|)$ candidates are retained:

$$\mathcal{A}^* \;=\; \operatorname*{arg\,max}_{\mathcal{B} \subseteq \mathcal{V},\, |\mathcal{B}| = K} \sum_{\alpha^{(k)} \in \mathcal{B}} S_k. \tag{15}$$

This selection and IC orientation remain adaptive reuse of trailing data. They do not demonstrate factor discovery or alpha profitability. Separate train-fit fixed/dynamic/random numerical controls are reported in Section 6.

Causal scope. The native row-wise causal gate fails. The native `delay().bfill()`, whole-Series normalisation and full-frame volume-availability branch can change earlier rows when later rows in the supplied prefix change. Synthetic probes that alter prices after the outer cutoff leave outputs unchanged, but establish only outer-cutoff isolation. They do not certify causality of rows inside the prefix or news availability. Historical LLM outcomes are consequently interpreted as observations under this native construction, not as outputs of a retrospectively repaired causal selector.

Custom handler formulas are in Appendix C. Native BFE multiplies $(\%B - 0.5)$ by signed $VolZ_5$ before tanh: negative volume z-score reverses the sign and zero cancels it. There is no squeeze/volume-expansion gate. Registry/prototype preprocessing differ, and a selected display rank is not a permanent handler ID. Six metadata slots have description/handler mismatches, without the literal wrong formula in the affected prompts; their separate semantic effect is unmeasured.

3.5. Sentiment Signal Construction

Let $\mathcal{D}_e = \{d_1, \ldots, d_{|\mathcal{D}_e|}\}$ denote the target-stock articles admitted by the date, identity and scorer-provenance rules at cutoff τ_d, $d = e - 1$. Each article has score

$$\rho_i \;=\; \mathrm{Scorer}_i(d_i) \;\in\; [-1,\, 1], \tag{16}$$

For the pinned classifier, $\rho_i = P_i(POS) - P_i(NEG)$ with NEG/POS/NEU label indices 0/1/2. The scorer is `5CD-AI/Vietnamese-Sentiment-visobert`, distinct from the ViSoBERT backbone [12, 13]; its local revision is `d7738b85200272f10ba8b048d80236c8c582b581`. The saved scoring recipe uses 256-token truncation and batches of eight. This identifies a scoring procedure, not financial-domain validation. Unknown legacy scorer identities are not inferred from score values.

The unweighted corpus statistic, distinct from the weighted context delivered to Decision, is

$$\bar{\rho}_e \;=\; \frac{1}{|\mathcal{D}_e|} \sum_{i=1}^{|\mathcal{D}_e|} \rho_i. \tag{17}$$

For a nonempty admitted corpus, define article age in days and the time-decayed context. The final line specifies the application's relative-peer extension when a valid peer set $\mathcal{R}_e$ exists:

$$\Delta_i(\tau_d) = \tau_d - \mathrm{date}(d_i), \tag{18}$$

$$w_i(\tau_d) = 2^{-\Delta_i(\tau_d)/h}, \qquad h = 30 \text{ days}, \tag{19}$$

$$S_{\mathrm{decay},e} = \frac{\sum_{d_i \in \mathcal{D}_e} w_i(\tau_d)\rho_i}{\sum_{d_i \in \mathcal{D}_e} w_i(\tau_d)}, \tag{20}$$

$$\mathrm{Rel}_{\mathrm{sent},e} \;=\; S_{\mathrm{decay},e} \;-\; \frac{1}{|\mathcal{R}_e|} \sum_{s' \in \mathcal{R}_e} S^{(s')}_{\mathrm{decay},e}, \tag{21}$$

The half-life is $h = 30$ days. The peer comparison requires at least three valid dated-and-scored articles per admitted company. It is *not an observed peer signal in this study*: the frozen provider returns an aggregate/count report with empty article-detail and peer lists. The normalizer's relative and z-score fields are therefore zero-valued placeholders on that path, not the peer contrast in Eq. 21. We do not use that placeholder as measured neutral peer information.

Batch definition and low-news stability. The target window spans 90 days and requires at least three distinct identities with accepted scorer provenance and finite scores. Missing context is not an observed neutral zero. The separate normalizer reliability flag requires at least eight articles; a false flag alone does not mean that the aggregate was missing. Explicit availability times are retained; date-only publication is conservatively available the next day and undated records are rejected. These rules cannot authenticate original content availability retrospectively.

Legacy inventory contains 180 article records and 177 identities. All 180 legacy scorer identities are unknown, and no legacy cutoff meets the strict three-valid-article rule; old runtime counts are distinct from this eligibility audit. The reconstructed study and 154 locally rescored excerpts have separate provenance. Without independent financial-news labels or authenticated historical content, neither financial-domain accuracy nor point-in-time-certified sentiment is established. Neither the weighted aggregate nor the peer extension enters native OHLCV factor formulas or their ranking.

3.6. Multi-Agent Decision Function

The system decomposes $\mathcal{F}$ into the following role-level functions, with the history prefix explicitly distinguished from the chart window:

$$\mathbf{r}^{\text{ind}} = \mathcal{F}_{\text{ind}}\Big(\mathbf{X}_e\Big), \tag{22}$$

$$\mathbf{a}_e = \mathcal{F}_{\text{factor}}\Big(\mathbf{X}_e^{\text{hist}},\ \mathcal{A}^*\Big), \tag{23}$$

$$\mathbf{c}_e^{\text{sent}} = \mathcal{F}_{\text{context}}\Big(S_{\text{decay},e},\ n_e,\ m_e\Big), \tag{24}$$

$$\mathbf{r}^{\text{alpha}} = \mathcal{F}_{\text{text}}\Big(\mathbf{a}_e,\ \mathbf{c}_e^{\text{sent}}\Big), \tag{25}$$

$$\mathbf{r}^{\text{pat}} = \mathcal{F}_{\text{pat}}\Big(\mathbf{I}_e^{\text{kline}}\Big), \tag{26}$$

$$\mathbf{r}^{\text{trend}} = \mathcal{F}_{\text{trend}}\Big(\mathbf{I}_e^{\text{trend}}\Big), \tag{27}$$

Here $\mathbf{I}_e^{\text{kline}}$ and $\mathbf{I}_e^{\text{trend}}$ are the two chart inputs, n_e is the admitted article count and m_e carries context availability/provenance. In the evaluated backtest, $\mathcal{F}_{\text{text}}$ formats the factor/sentiment reports without an extra Alpha reasoning-model call; Indicator is also deterministic. Pattern and Trend use their recorded vision requests. The relative-peer extension above is not supplied as an observed value.

Decision aggregates the role reports through prompt instructions:

$$\hat{y}_{e,L} = \mathcal{F}_{\text{dec}}\Big(\mathbf{r}^{\text{ind}},\ \mathbf{r}^{\text{alpha}},\ \mathbf{r}^{\text{pat}},\ \mathbf{r}^{\text{trend}}\Big) \in \{-1, +1\}. \tag{28}$$

The inclusion of factor and sentiment sections is variant-specific: Full includes both, Base neither, A only factors, S only sentiment, and P the predetermined past-donor factor replacement with current sentiment. Thus $\mathbf{r}^{\text{alpha}}$ denotes enabled quantitative/text context, not evidence of a learned fusion model. General priority instructions can remain even when their corresponding sections are omitted. Compact main/upstream and explicit-five-row factorial families are reported separately. Binary prediction is distinct from the account action: SHORT maps to CASH; LONG enters the specified cycle under the separate execution assumptions.

3.7. Walk-Forward Evaluation Protocol

Valid boundaries satisfy $W \le e \le T-L$. For each security, the study evaluates the chronologically ordered eligible dates in its fixed schedule:

$$\mathcal{T} = \Big\{ e_i \mid d_i = e_i - 1 \in \mathcal{D}_{\text{test}},\ W \le e_i \le T - L \Big\}, \qquad e_1 < \cdots < e_{N_s}, \tag{29}$$

where $\mathcal{D}_{\text{test}}$ comprises two selected dates in each of six blocks, with $N_s = 12$ eligible units for eight securities and six for BHN. This is not a continuous uniform $S = 3$, twenty-point backtest. Train and validation labels are purged at the next partition boundary. The pseudocode describes the implemented input/report comparison, while the causal qualifications above remain applicable.

Directional accuracy, its paired difference, the reference cycle return and wealth are

$$\text{Acc} = \frac{1}{|\mathcal{T}|} \sum_{e \in \mathcal{T}} \mathbf{1}\big[\hat{y}_{e,L} = y_{e,L}\big], \tag{30}$$

$$\Delta\text{Acc} = \text{Acc}_{\text{Full}} - \text{Acc}_{\text{Base}}, \tag{31}$$

$$q_e = a_e \left[\frac{C_{e-1+L}}{O_e} \frac{(1 - c_{\text{slip}})(1 - c_{\text{fee}})}{(1 + c_{\text{slip}})(1 + c_{\text{fee}})} - 1 \right], \tag{32}$$

$$W_{k+1} = W_k \, (1 + q_{e_k}), \qquad W_0 = 1, \tag{33}$$

$$R_{\text{acct}} = \frac{W_K}{W_0} - 1, \tag{34}$$

The paired ΔAcc is a descriptive Full–Base bundle difference, not an isolated alpha treatment. BA is the primary class-aware metric and is defined with MCC in Section 5. In the fractional reference ledger, $a_e = 1$ for LONG and zero for SHORT/CASH. Each LONG pays fee and slippage at next Open and target Close; $W_0 = 1$ is normalized wealth, equivalent in return to the VND50 million account initialization. Cycles do not overlap on the selected schedule. Separate accounts are not pooled portfolio capital, and settlement/capacity sensitivities retain receivables and residual shares rather than assuming the reference product always applies.

For daily-close wealth W_t^{day} including CASH gaps and the initial anchor, let $r_t^{\text{day}} = W_t^{\text{day}} / W_{t-1}^{\text{day}} - 1$. The account risk conventions are

$$\text{Sharpe}_{\text{acct}} = \sqrt{252} \, \frac{\mathbb{E}[r_t^{\text{day}}]}{\text{Std}[r_t^{\text{day}}]}, \tag{35}$$

$$\text{MDD}_{\text{acct}} = \max_{0 \le t \le T_{\text{day}}} \left(1 - \frac{W_t^{\text{day}}}{\max_{0 \le j \le t} W_j^{\text{day}}} \right). \tag{36}$$

The daily Sharpe uses population SD, zero risk-free rate and a 252-observed-session convention; it is undefined if dispersion is zero or insufficient. Annualised cycle ratios are not reported for the irregular/gapped schedule. These daily conventions do not certify exchange calendars, executable returns or reliable sparse-trade risk estimates.

Finally, with $\mathcal{X} = \{e_k : a_{e_k} = 1\}$, trade-level summaries are

$$\text{HitRate} = \frac{1}{|\mathcal{X}|} \sum_{e \in \mathcal{X}} \mathbf{1}[q_e > 0], \qquad \text{AvgTrade} = \frac{1}{|\mathcal{X}|} \sum_{e \in \mathcal{X}} q_e, \tag{37}$$

Both are undefined when no trade is executed, recorded as NA/null in research reporting rather than zero. CASH cycles do not dilute the executed-trade denominator. Daily-close maximum drawdown includes the initial wealth anchor, and fee/slippage levels are research assumptions, not authenticated realised venue costs.

4. System architecture

4.1. Overview

The implementation uses a LangGraph directed graph with five specialised roles and a shared typed state. Indicator, Pattern, Trend and Decision roles, shared state, chart tools and trendline procedures reuse or modify QuantAgent components; the upstream MIT notice is preserved in the repository [1]. Alpha adds numerical factor and Vietnamese sentiment reports. This integration is an engineering adaptation, distinct from inventing the inherited graph or validating a learned knowledge-fusion method.

Figure 1 shows the actual information flow. In the evaluated backtest, Indicator is deterministic and the numerical/report Alpha path bypasses an additional alpha-reasoning LLM call. Pattern and Trend consume their separate chart images, while Decision receives the condensed reports and enabled quantitative/text sections. The five role names consequently do not imply five hosted LLM calls per forecast. Model aliases and actual request accounting are specified in Section 5.

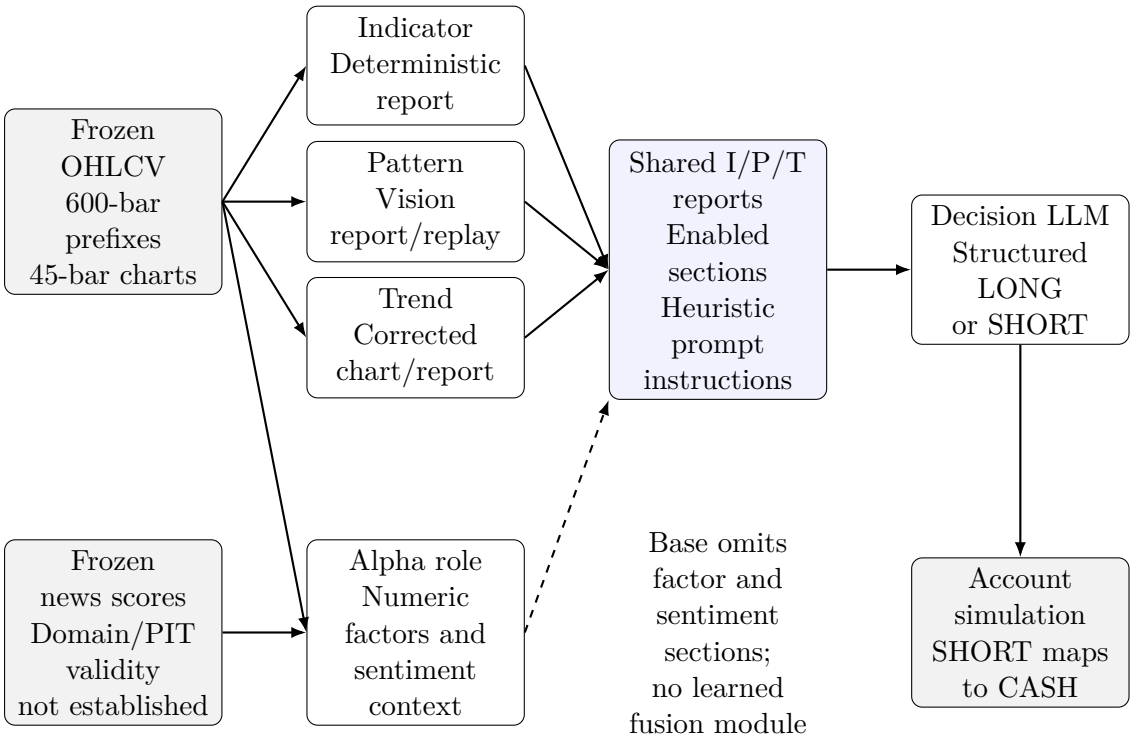


Figure 1: System pipeline with shared-report Full/Base comparison. Indicator and alpha backtest reports are deterministic/numerical; Pattern and Trend use vision requests. Decision aggregates enabled reports by prompt instructions.

4.2. Indicator Agent

Indicator Agent computes and renders five deterministic TA-Lib-based votes: MACD(12,26,9), RSI(14), Stochastic(14,3,3), Williams %R(14) and ROC(10). They are momentum/trend indicators, not five volatility oscillators. Its backtest node returns a report and messages, without calling an LLM or forwarding the declared state schema's complete numeric arrays. Bollinger variables can be computed on the alpha path; BB/ADX state declarations do not create two additional Indicator votes. Appendix D specifies the ordered boundaries and missing-data convention [16, 17].

Four-decimal rounding precedes classification, nonfinite samples are removed and zero is retained. Crossovers take priority where specified; overlapping RSI/Williams thresholds require ordered

branches rather than two simultaneous unconditional rules. Each missing/error result contributes a neutral vote within the fixed denominator of five. Directional plurality and the confidence categories are deterministic summary heuristics. They are not calibrated probabilities, and an immutable signal tally does not make a downstream narrative correct.

4.3. Alpha Agent

Alpha selection. The Alpha role evaluates 85 native candidates: 80 retained WorldQuant adaptations and five custom formulas. It supplies a selected top-five report from the trailing prefix, using the diagnostic score in Eq. 14. The transformations, metric support, IC orientation and native row-wise causal failure are defined in Section 3. A rolling own-history rank is not a cross-sectional rank, and the 21 exclusions are not all justified by an inability to translate cross-sectional operators. Full expressions and exception handling remain in the registry; Appendix C gives the custom formulas needed to read the reports.

Selection, IC sign and candidate diagnostics reuse trailing observations. Hence the current value is an input to a report, while the reported skill fields are in-sample ranking statistics. Neither choosing the top five nor reversing negative IC establishes out-of-sample profitability. Six selected metadata slots have description/handler mismatches; actual prompts, metadata and implementation identity are kept distinct. No historical prediction is repaired to agree with a corrected description.

Sentiment integration. Vietnamese sentiment is a separate contextual report, not an input to the native OHLCV factor formulas or score. The frozen provider supplies aggregate/count information and empty article-detail/peer lists. Relative sentiment and z-score fields on that path are consequently zero-valued outputs of the normalizer, rather than measured contemporaneous peer evidence. The three-article availability rule differs from the eight-article reliability flag. The 90-day window, 30-day decay, scorer revision and provenance limitations are specified in Section 3.

Figure 2 separates OHLCV factor selection from sentiment-context construction and shows the information delivered by the evaluated backtest. Optional live endpoints, lexicon fallbacks and peer discovery are repository capabilities; they are not evidence that the corresponding inputs were used or historically available in the evaluated calls.

4.4. Pattern Agent

Pattern Agent analyses a candlestick PNG rendered from the exact 45-bar window. The prompt requests a chart formation, recent-candle evidence, directional bias, confidence and trading implication. The report parser retains the structured text fields; format completion and consistency checks are distinguished from financial correctness. Figure 3 presents an actual archived chart pair rather than a synthetic demonstration.

Candle bodies are green when Close exceeds Open and red otherwise, including a doji. This is an Open–Close body rule, not the Vietnamese exchange board's reference-price colouring rule. The exchange-board convention is not used to infer candle direction. After inference, consistency rules compare the claimed bias with simple OHLCV-derived properties such as the final body and recent closes. A contradiction flag can reduce displayed confidence; it cannot reveal which candle the VLM actually attended to. Alternative text-only paths in the general application are not assumed to have replaced any frozen vision input.

4.5. Trend Agent

Trend Agent receives a separate chart with fitted support/resistance lines. The inherited fitting utility starts with a least-squares close-price line, selects upper/lower deviation pivots, and optimizes

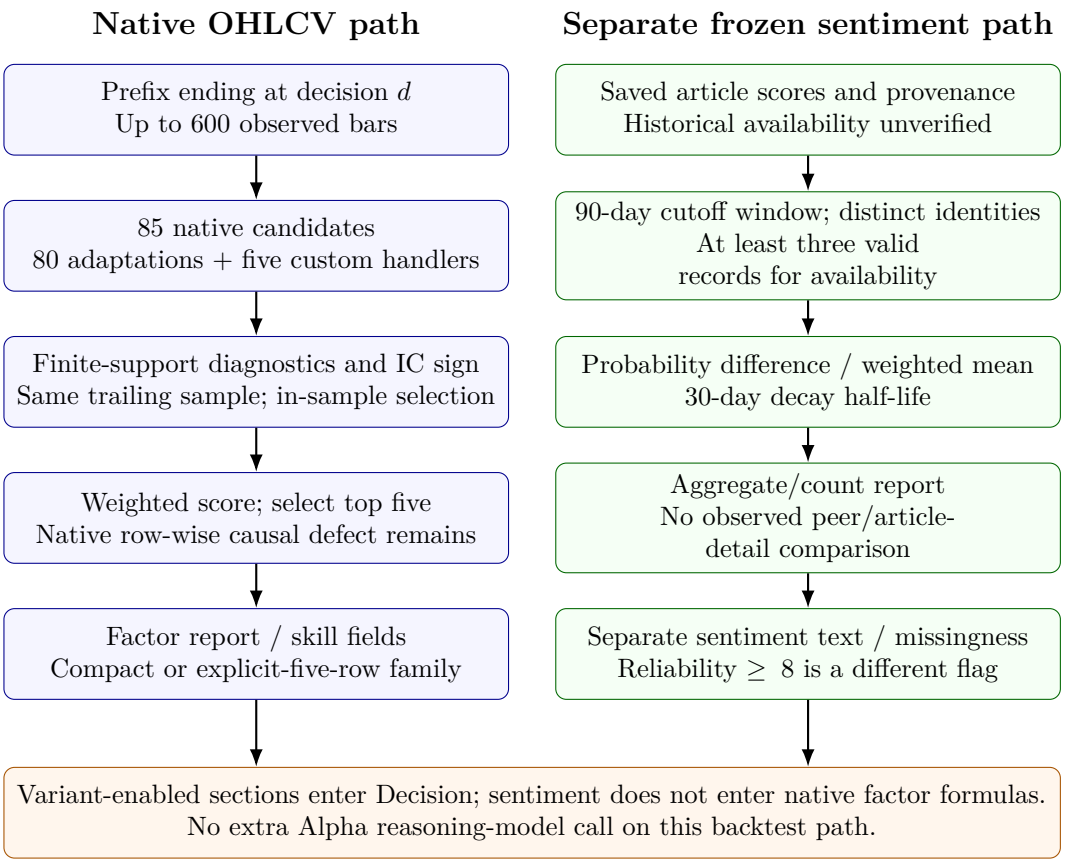


Figure 2: Alpha Agent: native OHLCV selection and separate sentiment context. Native factor diagnostics retain their causal/selection defects; the frozen sentiment path has no authenticated peer comparison. Both reports enter Decision only when enabled by the variant.

a slope through each pivot under a non-crossing constraint. It implements both a close-only channel and a high/low channel, the latter using close-price regression to find pivots and high/low data for the bounds. These are numerical lines over the supplied window, not human-verified latent support levels.

For row position u, a fitted line has form $b + mu$. A support bound must remain below the corresponding price sequence within the implementation tolerance; resistance must remain above it. The local slope search reduces its step when the squared-deviation objective fails to improve or violates the constraint. The VLM then describes direction, levels, slope, price relative to support, forecast and confidence. Support is drawn blue and resistance red. A deterministic consistency check against simple price/trend summaries does not validate every reported level or coordinate.

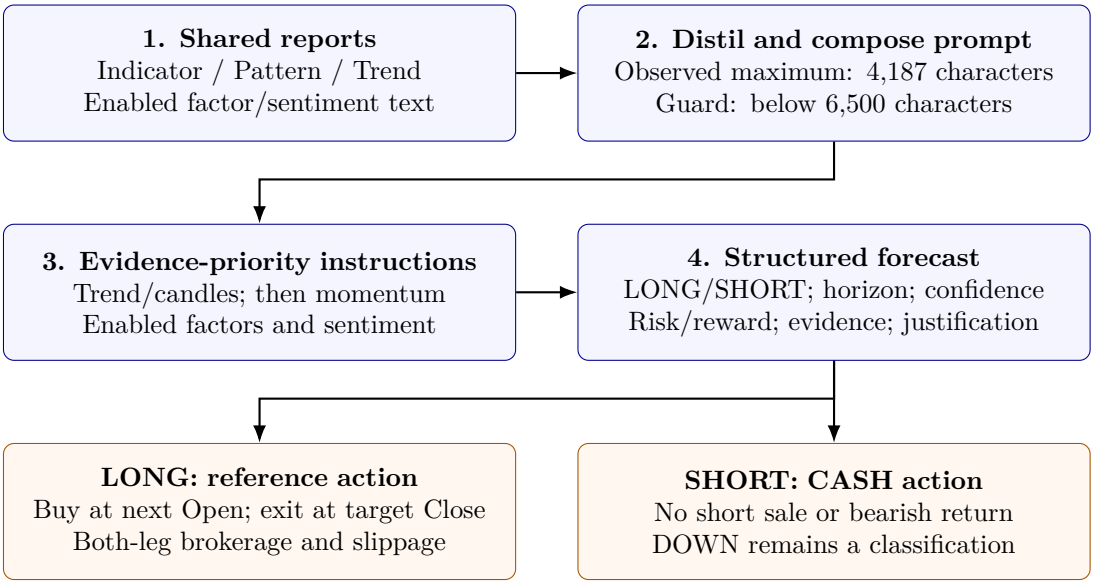
The renderer audit matched 204 images across 102 units byte for byte and pixel for pixel, and 240 actual vision requests matched their image caches and Vietnamese templates. It verified body geometry, colour and legend handles, not the model's visual understanding. In particular, the Vietnamese word *xanh* is less colour-specific than the corrected English "blue" wording. Inherited high-frequency/K-line prompt language also remains distinguishable from this daily-bar application. Matched local-pattern interventions and trend swaps would be needed to test whether the model uses the purported visual evidence.



Figure 3: Actual Pattern and corrected Trend inputs for CMG on 3 July 2023, the first chronologically eligible main unit. Bodies compare Close with Open; the Trend legend maps blue to support and red to resistance. The unit was selected by date then symbol/ID without inspecting outcomes.

4.6. Decision Agent

Decision receives condensed Indicator, Pattern and Trend reports plus enabled factor/sentiment sections. Reports are distilled and capped before prompt composition. The observed successful Decision prompts are at most 4,187 characters, below the 6,500-character guard. Figure 4 shows report composition, prompt instructions, structured output and the separate action mapping.



Prompt precedence and model confidence are heuristics; they do not certify reasoning, calibration, executable fills or a learned fusion mechanism.

Figure 4: Decision Agent pipeline. Evidence priorities are prompt heuristics rather than learned/calibrated weights. The structured forecast is evaluated separately from the conditional long-only account action.

The instructions ask for trend/key-level assessment, recent-candle evidence, momentum confirmation and comparison with enabled factor/sentiment context. Price/chart evidence has priority, quantitative factors may challenge it, and sentiment has lower priority. These instructions describe an intended analysis sequence; they are not an independently observed internal reasoning trace or a guaranteed conflict-resolution algorithm.

The required output includes a binary LONG/SHORT forecast, forecast horizon, confidence, risk/reward field, evidence for/against and justification. Model confidence and risk/reward statements do not constitute calibrated uncertainty or executable price levels. In the long-only reference ledger, SHORT maps to CASH and LONG to the specified next-open/target-close cycle. The classifier's labels and account execution remain distinct.

Base retains the same upstream content and general instructions but removes both alpha and sentiment sections; those concepts may still appear in general instructions. Thus Full–Base changes a bundle and prompt length, not solely one numerical factor. Main/upstream prompts use compact factor text, whereas factorial A/AS/P deliver five explicit rows. The resulting representation difference remains part of the interpretation of the separate panels.

5. Experimental Setup

5.1. Dataset and Point-in-Time Data Contract

The daily universe spans information technology (FPT, CMG), banking (VCB, MBB), retail (MWG), food/beverages (VNM, BHN) and aviation/tourism (HVN, VJC). This is a purposive, non-representative sample: BHN, CMG, FPT, HVN, MBB, MWG, VCB, VJC and VNM. The reconstructed vnstock/VCI price snapshot covers observed vendor sessions from 2020 through September 2026. It is not an authenticated submission-time snapshot. Contradictory BHN OHLC data trigger the frozen valid-suffix policy; insufficient subsequent warm-up excludes BHN from three early blocks. No price is corrected to improve a result. This leaves 102 eligible cells of the 12-date by nine-symbol grid in Figure 5.

5.2. Walk-Forward Protocol

Six blocks cover 2023 H2, 2024 H2, 2025 H1, 2025 Q4, 2026 Q2 and 2026 Q3, with two scheduled dates per block. These are twelve selected dates, not a continuous multi-year daily backtest. Cross-symbol observations share calendar shocks. The 51 eligible symbol–block scopes contain chronological train/validation/test partitions: every train label exits before validation starts, and every validation label exits before the first test decision. Cutoffs require the frozen history length; label spans touching the next partition are purged.

The reduced scope was amended after some inference to complete a bounded historical study. Its 375 registered Decisions include 193 explicitly reused successful executions and 182 further successes. The factorial subset was selected by metadata (symbol/date round-robin), not prediction quality, but this does not make the overall amendment preregistered. The larger legacy grids remain incomplete/superseded. Table 1 distinguishes shared r0 calls, additional realizations and the separate factorial panel; their observations are not summed as independent markets.

Table 1: Evaluation scopes and prompt variants in the frozen historical study. Shared main/upstream executions are counted once.

A. Recorded supports

Scope	N/exec.	Symbols	Dates	Blocks	UP/DOWN	Exec.	Decisions
Main	102	9	12	6	25/77	1	204
Upstream	9	9	3	3	2/7	3	54
Factorial	9	8	4	3	2/7	3	135

B. Prompt sections

Variant	Factors	Sentiment	Representation
B	–	–	Shared price reports
A	Current	–	Five-row factorial
S	–	Current	Factorial
AS	Current	Current	Compact main / five-row factorial
P	Past donor	Current	Five-row factorial

Main uses B/AS; upstream uses B/AS; factorial uses B/A/S/AS/P. Class counts are per execution. Upstream includes 18 main r0 calls: 204+54+135-18=375 unique Decisions. P preserves current names/schema/missingness and sentiment, replacing factor values/signals/skill metadata by one past train-only rank-vector donor.

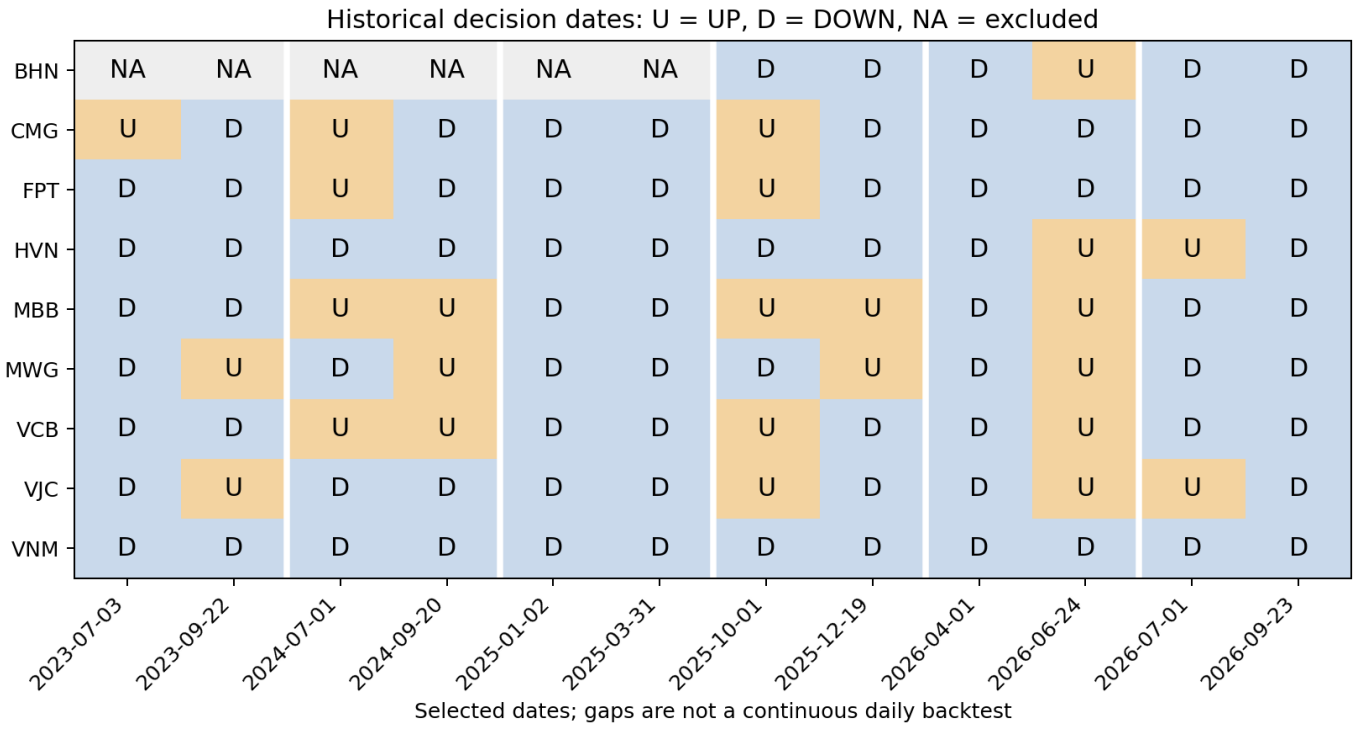


Figure 5: Selected historical decision dates and eligibility across nine equities. Missing BHN cells remain excluded rather than being filled with CASH predictions. Labels are shared evaluation truth, not a visualization selected from Full's successful forecasts.

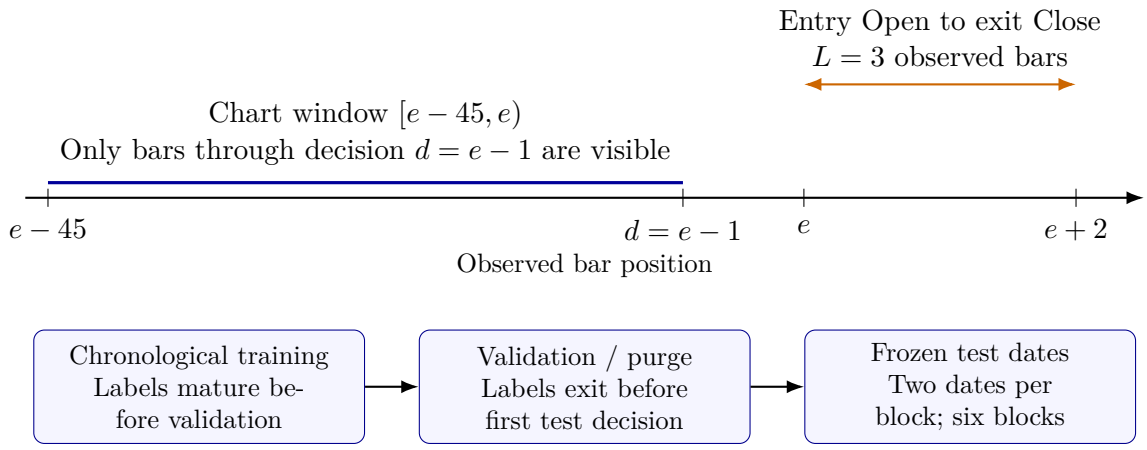


Figure 6: Walk-forward time-index and split contract. The 45-bar chart ends at $d = e-1$, entry is at e and the $L = 3$ target is at $e+2$. The final study visits twelve selected dates across six blocks; it does not use a uniform $S = 3$, twenty-point grid. Outer cutoff isolation does not certify native row-wise causality.

5.3. Paired Baseline and Disentangled Ablations

All nine classifiers use the same 102 main units and net target. CASH and always-LONG are constant predictions; always-LONG closes and pays costs every cycle. Train-majority uses purged training labels, choosing CASH on a tie. Momentum predicts LONG for a strictly positive trailing 20-session return. Logistic-price uses numerical price features; logistic-matched adds the recorded factor/sentiment/missingness inputs. Imputation, scaling, regularization and decision threshold selection follow chronological train/validation rules; final logistic refits use matured train plus validation only. We reuse their saved predictions rather than refitting after observing outcomes. The fixed factor-vote comparator uses training-fitted medians/orientation and differs from the native top-five policy and the auxiliary numeric-control study.

Matched numerical inputs do not equal chart images, report wording or pretrained knowledge. Full–Base differences likewise include prompt structure, report length and propensity to choose CASH. The upstream panel has nine units, B/AS and three regenerated upstream realizations with a Decision execution per realization. The factorial panel has nine different-support units, B/A/S/AS/P and three Decision executions with fixed upstream. Both have two UP/seven DOWN per execution, but only the former covers nine symbols. We do not infer main-grid variability or separate upstream/Decision variance components from their SDs.

P retains current factor names, headings, schema/missingness and sentiment, replacing scalar values, signals and skill fields using a predetermined earlier train-only rank-vector donor. Donor label exits precede validation/test. Rank mapping is not name-matched covariance preservation, a placebo distribution or exposure matching. Five explicit rows reach each A/AS/P factorial prompt. Main/upstream compact prompts are a separate representation family. Interaction is defined as $AS - A - S + B$; the observed best-single gap $AS - \max(A, S)$ is a different, post-hoc diagnostic and not an eighth factorial contrast.

5.4. Account and Statistical Metrics

BA averages UP and DOWN recall; MCC and both-class precision/recall accompany ordinary accuracy and LONG rate. Pooled metrics retain actual class supports. Symbol macro means are unweighted and report planned/defined denominators. Missing class support or a degenerate MCC denominator yields NA, not zero. Main contrasts use 10,000 paired whole-block resamples, seed 20261102, retaining dates, cross-section, variants and eligibility together. Repeat diagnostics jointly resample recorded blocks and paired execution indices. Intervals condition on finite draws; undefined counts remain visible. Circular block lengths, leave-one-block-out and Bartlett HAC at lags 0/1/2 are sensitivity diagnostics. HAC uses calendar-aligned accuracy differences in fixed-universe panels, with a $t(n_{dates} - 1)$ reference; few clusters, irregular gaps and post-inference additions prevent calibrated significance or power claims.

To specify the classification estimand, let TP/FN count UP outcomes and TN/FP count DOWN outcomes. We use

$$\begin{aligned} BA &= \frac{1}{2}\left(\frac{TP}{TP+FN} + \frac{TN}{TN+FP}\right), \\ MCC &= \frac{TP\,TN - FP\,FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}. \end{aligned} \tag{38}$$

The relevant metric is undefined when a required denominator is zero. Macro means average defined symbol-level values while retaining the planned and defined denominators. A CASH rule can have high ordinary accuracy in a DOWN-heavy sample without discriminating the two classes.

Each of 51 symbol–block accounts starts with VND50 million, without cross-symbol capital competition. LONG follows Eq. 4; CASH stays flat. Continuous hold buys once from the first entry through the last scheduled exit, including gaps, and is not a tenth classifier. Wealth includes pending receivables and residual shares at recorded mark-to-market prices. Daily-close drawdown uses an initial-capital anchor. Where supplied in companions, daily Sharpe uses population SD and a 252-session convention with zero risk-free rate; the retained Sortino convention uses SD of negative daily observations, not downside RMS. Annualised cycle ratios are NA for the irregular/gapped main schedules, and are not presented as investment-quality estimates from a handful of trades.

The reference assumes fractional fills at recorded prices and two-leg fees/slippage. Shares are assumed sellable in the afternoon of T+2 conditional on allocation and fill; sale proceeds are pending until the specified settlement phase. This convention follows the archived settlement source [18]. L=3 is a fixed observed-bar protocol choice, not an optimized horizon. Daily OHLC cannot authenticate opening-auction queues, price-limit fills, partial fills or correct volume units. All twelve sensitivities, including pending cash, advances, cost changes, sell charge, no-fill/delay proxies and lot/capacity assumptions, remain in Appendix F. Venue/calendar review flags, including ten HVN unit flags, are retained rather than converted to certified fills or synthetic CASH.

5.5. Implementation and Robustness Design

Archived SDK aliases are `openai/gpt-oss-20b` for text and `qwen/qwen3.8-27b` for vision. Configured temperature zero appears as 10^{-8} in observed SDK metadata; no hosted decoding seed is recorded. Text/vision output caps are 2,048/1,024 tokens. Retry/checkpoint records preserve individual attempts and completed outputs. Aliases and 40 response fingerprints do not authenticate an immutable weight revision or historical provider catalogue; 91 original Pattern requests lack epoch/provider-org fields and remain unknown. Offline replay reproduces saved-prediction metrics, not a fresh hosted inference experiment.

The closure contains 649 attempts, 615 successes and 34 errors, including 120 original Pattern calls replayed as shared dependencies. Observed usage totals 1,063,328 prompt and 340,736 completion tokens. Error usage/latency and monetary billing are unknown. Pattern tokens/latency describe original calls, not replay lookup cost; successful callback elapsed times are not wall-clock runtime. Full accounting, pins and hashes are in repository artifacts rather than an implementation appendix.

6. Backtest Results

6.1. Nine-Symbol Paired Benchmark

6.1.1. Main class-aware comparison

Table 2 uses all 102 main units, not the 180-point legacy study. Full accuracy is 65.69%, Base 60.78%, and CASH/train-majority 75.49%. Because 77/102 outcomes are DOWN, ordinary accuracy alone favours conservatism. Full BA is 52.96%, Base 49.71%, and logistic-price 55.61%; the Full–Base BA difference is 3.25 percentage points. All comparators, including less favourable ones, remain in the table. Undefined constant-rule MCC is NA.

Full and Base each identify seven of 25 UP outcomes. Full issues 24 LONG forecasts, with 17 false positives, versus Base's 29 LONG forecasts and 22 false positives. Thus the observed BA difference accompanies improved DOWN recall rather than additional captured UP cases. This pattern is compatible with a shift towards CASH; it does not identify why the shift occurred or establish information-specific alpha. Neither a higher score than Base nor a smaller loss than an always-invested rule is sufficient evidence of profitability or broad superiority.

Pooled and macro values are different estimands. BHN contributes six eligible units, the other eight symbols twelve; missing class support reduces some macro denominators. Table F.1 reports Full/Base by symbol. Companion S1 includes all 81 symbol–classifier rows, both-class metrics and all pooled/macro denominators, so comparator reporting is not restricted to selected symbols or the main table's aggregate.

Table 2: Class-aware performance on the same 102 historical market units. Macro summaries weight symbols equally and retain undefined denominators.

A. Pooled (percent except MCC)

Strategy	Acc.	BA	MCC	P_U	R_U	P_D	R_D	LONG
Full	65.69	52.96	0.060	29.17	28.00	76.92	77.92	23.53
Base	60.78	49.71	-0.005	24.14	28.00	75.34	71.43	28.43
CASH	75.49	50.00	NA	NA	0.00	75.49	100.00	0.00
Train-majority	75.49	50.00	NA	NA	0.00	75.49	100.00	0.00
Always-LONG	24.51	50.00	NA	24.51	100.00	NA	0.00	100.00
Momentum-20	49.02	47.32	-0.046	22.45	44.00	73.58	50.65	48.04
Logistic-price	67.65	55.61	0.114	33.33	32.00	78.21	79.22	23.53
Logistic-matched	60.78	52.42	0.044	27.27	36.00	76.81	68.83	32.35
Factor vote	51.96	51.97	0.034	26.00	52.00	76.92	51.95	49.02

B. Unweighted symbol macro summaries

Strategy	BA (%)	Defined	MCC	Defined
Full	55.38	8/9	0.08	8/9
Base	51.48	8/9	0.00	8/9
CASH	50.00	8/9	NA	0/9
Train-majority	50.00	8/9	NA	0/9
Always-LONG	50.00	8/9	NA	0/9
Momentum-20	50.67	8/9	-0.01	8/9
Logistic-price	59.00	8/9	0.17	7/9
Logistic-matched	56.55	8/9	0.08	8/9
Factor vote	46.86	8/9	-0.04	8/9

$N = 102$, UP/DOWN=25/77. P/R denote class precision/recall. NA is undefined, including MCC for constant predictions. Train-majority is fit on purged train labels. Logistic-matched uses common numerical market inputs, not identical chart/text representations. Full per-symbol/class metrics are supplied in companion S1; hold has no classification prediction.

6.1.2. Account outcomes and exposure

Full mean net account return is -0.797%, versus Base -0.750% and CASH zero (Table 3). Full therefore does not even outperform Base on this main economic mean, despite its higher BA. Always-LONG cycles and continuous hold have different costs and gap exposure, and neither is substituted for the other. The held-session fractions include the long CASH gaps between the two cutoffs in each block; their low values should not be confused with LONG percentages over scheduled decision units.

Returns and drawdowns summarize separate accounts, not one portfolio, total-return index or certified executable strategy. Risk ratios cannot recover evidence of profitability from sparse trades.

The main schedules have 57 observed sessions between cutoffs, so pending-cash/on-demand-advance equivalence on these accounts provides little evidence about rapid capital reuse. All twelve economic sensitivities are reported in Table F.2. The all-entries-missed scenario gives zero but is not a mathematical worst-return bound; other assumptions may introduce unfilled entries, delays, pending balances or residual shares. Predictions and classification labels are unchanged across these sensitivities.

Table 3: Account outcomes under the fractional reference ledger. Means summarize 51 separate symbol–block accounts per strategy and do not establish executable portfolio returns.

Reference ledger

Strategy	Return (%)	MDD (%)	Held (%)	Orders
Full	-0.797	1.37	2.35	0.94
Base	-0.750	1.45	2.84	1.14
CASH	0.000	0.00	0.00	0.00
Train-majority	0.000	0.00	0.00	0.00
Always-LONG	-3.000	4.52	10.00	4.00
Momentum-20	-1.636	2.58	4.80	1.92
Logistic-price	-0.683	1.18	2.35	0.94
Logistic-matched	-1.105	1.60	3.24	1.29
Factor vote	-1.457	2.49	4.90	1.96
Continuous hold	-2.127	12.94	100.00	2.00

Each row averages 51 independent-capital symbol–block accounts, not a portfolio. Held is the fraction of observed calendar sessions held, including scheduled CASH gaps; it differs from the LONG rate. Continuous hold buys once and sells once across gaps. Dividends/corporate actions and fills are not certified. Annualised cycle Sharpe/Sortino are omitted for irregular/gapped schedules.

6.2. Disentangled Alpha–Sentiment Ablation

6.2.1. Repeated executions and factorial contrasts

Table 4 retains all three executions, sample SD, range and cutoff-level prediction disagreement summaries. The two panels have nine market units each but different symbols, dates and prompt families. Their means and SDs are not transferable to the broad 102-unit main study. Zero metric SD does not imply identical predictions: factorial S has unchanged aggregate BA but nonzero disagreement. Return means use nine upstream or eight factorial symbol–block accounts per variant/execution.

On the factorial panel the AS–B BA mean is -9.52 pp and AS–P is -11.90 pp. Interaction $AS - A - S + B$ is $+14.29$ pp, whereas the gap of mean BA between AS and the observed best single component is zero. These are different quantities. All seven descriptive BA family envelopes contain zero; only 6,986 of 10,000 BA draws retain both classes and 3,014 are undefined. Conditional intervals are not zero-imputed or advertised as calibrated familywise coverage. All factorial mean account returns are negative.

The upstream AS–B mean BA difference is $+4.76$ pp, with conditional percentile interval $[0, 10]$ pp. This higher-performing small panel does not replace the main or factorial outcomes. Actual shared reports and paired prompt contents match their frozen parents, but 40 of 147 groups have a successful-call order different from the listed job order, and nine span execution epochs. Routing/quota/resume did not implement full counterbalancing. We retain every outcome and disclose

the resulting time/order confounding rather than selecting a clean-looking subset or repairing it retrospectively.

Table 4: Execution variability and paired contrasts on two distinct nine-unit panels. Repeated executions do not add independent market dates.

A. Three executions: mean (sample SD)

Panel/variant	Acc. (%)	BA (%)	BA range	Disagr. (%)	Return (%)
U B	85.19 (16.97)	90.48 (10.91)	[78.57, 100.00]	22.22	0.37 (0.38)
U AS	92.59 (6.42)	95.24 (4.12)	[92.86, 100.00]	7.41	0.50 (0.22)
F B	44.44 (11.11)	28.57 (7.14)	[21.43, 35.71]	14.81	-1.70 (0.55)
F A	29.63 (6.42)	19.05 (4.12)	[14.29, 21.43]	22.22	-2.03 (0.31)
F S	22.22 (0.00)	14.29 (0.00)	[14.29, 14.29]	22.22	-2.34 (0.22)
F AS	29.63 (16.97)	19.05 (10.91)	[7.14, 28.57]	22.22	-2.23 (0.63)
F P	48.15 (6.42)	30.95 (4.12)	[28.57, 35.71]	14.81	-1.42 (0.33)

B. BA contrasts in percentage points

Contrast	Mean	SD	Range	Percentile	Family env.
U AS-B	4.76	8.25	[0.00, 14.29]	[0.00, 10.00]	NA
F AS-B	-9.52	10.91	[-21.43, 0.00]	[-22.22, 9.52]	[-37.14, 18.10]
F A-B	-9.52	4.12	[-14.29, -7.14]	[-20.37, 14.29]	[-37.14, 18.10]
F AS-S	4.76	10.91	[-7.14, 14.29]	[-6.67, 24.07]	[-22.86, 32.38]
F S-B	-14.29	7.14	[-21.43, -7.14]	[-29.63, 10.00]	[-41.90, 13.33]
F AS-A	0.00	12.37	[-14.29, 7.14]	[-20.00, 11.11]	[-27.62, 27.62]
F AS-P	-11.90	14.87	[-28.57, 0.00]	[-28.57, 4.76]	[-39.52, 15.71]
F interaction	14.29	14.29	[0.00, 28.57]	[-16.67, 38.89]	[-13.33, 41.90]

U: nine symbols/three dates/three blocks; F: eight symbols/four dates/three blocks. Each panel has nine units and UP/DOWN=2/7 per execution. Returns average nine U or eight F accounts per variant/execution. BA intervals retain 6,986 finite and 3,014 undefined draws of 10,000; envelopes are descriptive, not calibrated simultaneous coverage. U AS-B is separate from the seven F contrasts. Forty/147 groups deviate from listed variant order; nine span execution epochs.

6.3. Robustness and Sensitivity

6.3.1. Temporal and dependence diagnostics

Table 5 keeps the all-102 primary cohort and two overlapping structural cohorts. The primary Full–Base BA interval is $[-1.18, 8.13]$ pp. The common-eight cohort has 96 units over twelve dates; recent-nine has 54 over six dates. They are sensitivity views, not independent replications or grounds for choosing a more favourable headline. The six-block sample includes an all-DOWN early-2025 block: within-block BA without both classes is undefined rather than fabricated.

Past Close/SMA60 contexts distinguish bull, bear and sideways observations. These are per-security descriptions, not validated market-wide index regimes or a regime-conditioned trading policy. Additional volatility and turnover-proxy thresholds are fitted on purged training data, but their post-inference stratification remains descriptive. Close times Volume is in vendor units, not measured spread/depth liquidity. Exposure based on three target sessions differs from whole-calendar exposure with gaps.

Calendar-aligned HAC accuracy effects for common-eight/recent-nine are +3.125/ + 9.259 pp. At lags 0/1/2, common-eight SEs are 0.026042/0.023146/0.020172 with p-values 0.255346, 0.204107

and 0.149610; recent-nine SEs are 0.040713/0.047112/0.045710 with p-values 0.072051, 0.106554 and 0.098654. These are accuracy, not BA, effects on twelve/six selected dates. Companion S4 retains all six rows, the calendar masks, all regime strata, block-length sensitivities and leave-one-block-out diagnostics. Few clusters and irregular dates prevent reliable significance, null-effect or calibrated-coverage conclusions.

Table 5: Overlapping temporal cohorts and observed trend strata. Small-block intervals and calendar-aligned HAC diagnostics are descriptive rather than calibrated significance tests.

A. Cohorts: BA percent; difference/interval in pp

Cohort	N	Dates	Blocks	Full	Base	Δ	Interval
All	102	12	6	52.96	49.71	3.25	[-1.18, 8.13]
Common eight	96	12	6	50.69	48.61	2.08	[-1.20, 5.26]
Recent nine	54	6	3	53.08	46.67	6.41	[0.00, 12.50]

B. Observed per-security trend contexts (percent)

Trend	N	UP/DOWN	Full BA	Base BA	Full LONG	Base LONG
Bear	35	5/30	43.33	41.67	11.43	14.29
Bull	33	9/24	47.92	53.47	36.36	39.39
Sideways	34	11/23	59.49	46.25	23.53	32.35

Cohorts overlap and are not pooled as extra samples. Trend uses past Close/SMA60 with strict +/-2% thresholds; it is not an index-regime policy. Primary intervals retain both dates, full cross-sections and pairing within blocks. HAC concerns accuracy rather than BA and is reported for both fixed-universe panels at lags 0/1/2 in S4.

6.3.2. Auxiliary numeric selector controls

Table 6 is a separate numeric study, not an LLM factor-selector ablation. Causal adapters and train-only eligibility, imputation, scaling, orientation and ranking are fitted once per symbol–block scope, then retained through validation/test. Fixed custom five, dynamic train top-five, the same IDs without sign flip, and all twenty random eligible-five trials produce 2,346 predictions on the same 102 units. No random seed is selected after observing results.

Dynamic-minus-fixed, dynamic-minus-random-family and dynamic-minus-no-flip BA differences are respectively +15.84, +7.33 and +6.44 pp. Exact six-block two-sided sign-flip values are 0.0625, 0.125 and 0.53125; Holm correction over all 23 declared hypotheses gives adjusted $p = 1$ throughout. Unadjusted percentile intervals and the entire random dispersion remain visible. Positive point differences do not establish superiority, and the no-flip contrast does not identify sign-flip value inside the LLM pipeline.

Native diagnostics separately account for 102 cutoffs, 8,670 candidate opportunities and 510 selected slots. WQ42 occurs at 52/102 cutoffs (52/510 slots); adjacent nonzero IC signs flip 1,282 times over 7,894 eligible transition pairs. These are dependent in-sample diagnostics, not out-of-sample alpha stability. Complete per-factor frequency, eligible denominators, IC signs, windows and all 21 exclusions are retained in the registry companion. Numeric controls do not repair native row-wise noncausality, provide the missing fixed/random LLM experiment, or certify original-expression equivalence.

Table 6: Train-fit numeric selector controls using separate causal adapters. These auxiliary results neither repeat the native LLM selector experiment nor repair its row-wise causal defect.

A. Pooled numeric results; random mean (SD)

Strategy	Trials	Acc. (%)	BA (%)	MCC	Random BA range
Fixed custom five	1	49.02	39.22	-0.193	–
Dynamic train top-five	1	62.75	55.06	0.093	–
Same IDs, no flip	1	50.98	48.62	-0.024	–
Random eligible five	20	51.27 (5.32)	47.74 (5.38)	-0.039 (0.094)	[37.97, 57.17]

B. Three primary contrasts; multiplicity family = 23

Contrast	ΔBA (pp)	Percentile interval	Exact p	Holm p
Dynamic - fixed	15.84	[6.82, 25.22]	0.0625	1.0000
Dynamic - random mean	7.33	[0.57, 13.75]	0.1250	1.0000
Dynamic - no flip	6.44	[-10.93, 20.26]	0.5312	1.0000

$N = 102$, UP/DOWN=25/77, 51 train-fit scopes. Numeric adapters/preprocessing are separate from the native LLM pipeline. All 20 random trials and all 23 contrasts remain in S3. Exact tests have 64 six-block sign assignments; percentile intervals are unadjusted, Holm uses the full 23-hypothesis family. No contrast rejects at 0.05.

7. Limitations & Future Work

7.1. Limitations

What the comparisons establish. The archived implementation yields identifiable forecasts and replayable metrics, but its Full–Base contrast changes a bundle of inputs and prompt structure. Main/upstream compact reports differ from explicit factorial values. Matching report parents or numerical inputs does not match text/image representations, pretrained knowledge, exposure or semantic use. The rank-vector placebo tests one predetermined past replacement, not a distribution of independent placebo treatments. Consequently, these results describe recorded behaviour and do not isolate causal alpha, sentiment or selector value. Byte-level duplication in legacy artifacts is audited rather than treated as proof that historical branch labels were correct; those legacy ablations no longer support component attribution.

Causal construction and domain validity. The native row-wise causal failure in Section 3 remains unresolved in the frozen study. Its in-sample ranking/orientation and heuristic expression translations limit any factor-discovery interpretation. The numerical controls use a separate causal path, so their passing probes cannot certify the earlier native inputs. Similarly, saved scorer probabilities, a pinned local checkpoint and strict date rules cannot authenticate the original news content or publication availability. No independent financial-domain labels, agreement/adjudication or domain test performance are available. Missing sentiment is not evidence that sentiment was neutral, and rescoring is not validation. Repairing these defects for a new experiment would require versioned inputs and a new protocol, not rewriting old predictions.

Sampling, dependence and hosted variability. The universe is purposive and potentially survivor-biased; point-in-time listing/delisting/suspension membership is unverified. All outcome dates precede the November 2026 freeze, so model training contamination remains possible even if a named model was publicly released before some test dates. Twelve dates and six blocks, plus three executions on each small panel, do not establish generalization, effective sample size or calibrated uncertainty. Repeats, correlated symbols, overlapping cohorts and repeated bootstrap draws are

not extra independent markets. Order deviations, epoch/routing changes and unknown immutable served weights further confound distributional comparisons. The historical scope amendment makes this an exploratory study rather than an untouched confirmatory evaluation.

Economic and visual scope. Negative account means do not demonstrate a deployable strategy. Reference fills, fee/slippage, settlement phases, lot/capacity limits and volume units are assumptions or proxies, while corporate-action adjustments, price-limit queues, actual spread/depth and partial fills are unauthenticated. Separate account averaging omits portfolio capital competition. L=3 is not optimized. Visual byte/pixel/legend integrity and deterministic consistency flags do not show that the VLM read the claimed candle or trendline; matched counterfactual charts and local-evidence annotation would be needed to test that stronger claim. Prompt precedence is not a learned or guaranteed conflict-resolution mechanism.

7.2. Future Work

A prospective versioned study should repair native row causality, repeat selection/preprocessing inside chronological training, and preserve a later untouched evaluation interval. Candidate search can be assessed with procedures such as White's Reality Check or a Deflated Sharpe analysis [11, 10]; neither is claimed as completed here. Independent financial-news labels and human visual-evidence annotations are needed before domain/semantic reliability claims. More calendar coverage and controlled repeated hosted calls would be required for stable model/component conclusions; executable portfolio validation needs separately authenticated market and settlement information. Public release is also conditional on rights and a real archive identifier. These are outstanding requirements, not benefits inferred from the current experiment.

7.3. Broader Impact

The implementation is research decision support and should not be represented as profitable trading advice. A fluent report can make uncertain or contradictory evidence appear more reliable than it is, especially to readers without access to the input provenance or account assumptions. Separating the model's forecast/confidence from the action ledger and reporting negative results are therefore part of responsible interpretation. Financial-news and price data also carry redistribution constraints; an internal reproducibility archive is not permission to publish every source record. Prospective deployment would require independent data, domain and execution validation beyond the present historical study.

8. Conclusion

This study integrates quantitative alpha context and Vietnamese-language sentiment into a five-role LLM framework for Vietnamese-equity trading research. It specifies the single-asset factor transformations, separate sentiment construction, shared report composition and forecast-to-account action mapping. The common-support evaluation compares the full input bundle with simple numerical controls and complementary repeated component/placebo panels.

On 102 market units, Full balanced accuracy is 52.96% versus Base 49.71%, while a price-only logistic comparator reaches 55.61%. The Full–Base difference accompanies fewer false LONG forecasts and the same seven captured UP outcomes. Full's mean separate-account net return is -0.797%, compared with -0.750% for Base. Quantitative integration therefore changes recorded decisions, but this study does not establish stable predictive superiority, causal component value or deployable profitability. Native row-wise factor noncausality, historical news/domain uncertainty and limited

calendar/repeat support constrain interpretation. A prospective evaluation with corrected causal construction and independent domain/execution validation is needed to test generalisation.

CRediT authorship contribution statement

Van-Tang Nguyen: Conceptualization (C); Software (So); Validation (Va); Investigation (I); Original draft preparation (O); Project administration (P); Funding acquisition (Fu).

Hoang-Dung Phan: Methodology (M); Software (So); Formal analysis (Fo); Investigation (I); Resources (R); Data curation (D); Original draft preparation (O); Writing-review and editing (E); Visualization (Vi); Supervision (Su).

Thanh-Tam Do Thi: Methodology (M); Investigation (I); Data curation (D); Original draft preparation (O); Writing-review and editing (E); Supervision (Su).

Phuong-Thai Nguyen: Methodology (M); Investigation (I); Data curation (D); Original draft preparation (O); Visualization (Vi); Supervision (Su).

Hong-Viet Tran: Conceptualization (C); Methodology (M); Software (So); Validation (Va); Formal analysis (Fo); Data curation (D); Original draft preparation (O); Writing-review and editing (E); Project administration (P).

Declaration of competing interest

Authors state no conflict of interest.

Data and code availability

The project repository records the implementation and earlier result artifacts. The final reproducibility package and scientific companions S1–S4 have been prepared as a private local draft. Public distribution rights and a persistent archive identifier remain pending; no DOI or public availability of the full package is asserted. OHLCV/news snapshots are reconstructed research inputs rather than authenticated submission-time data. Distribution remains subject to source terms, news copyright, model licences and permissions; the upstream QuantAgent MIT notice does not grant rights over all additions or data.

Companion S1 contains `S1_classification.csv`, `S1_per_class.csv` and `S1_macro.csv`; S2 contains all executions, contrasts and disagreements; S3 contains numeric metrics, random trials and the full family; S4 contains calendar, HAC, block sensitivity, regimes and dependence diagnostics. Exact source paths and hashes are recorded in the manuscript release. The private raw package separately retains inputs, messages/images, responses/errors, parse/checkpoints, pins and provenance.

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Appendix A. Agent System Prompts

This appendix specifies the role-level prompt and output contracts used to interpret the system. The summaries below are English explanations of the actual Vietnamese prompt families, not new verbatim templates or translations claimed to be identical to each archived request. Exact messages, image payloads and request metadata remain in the companion archive. The evaluated path must be distinguished from optional prompts present in general live-mode code.

Appendix A.1. Decision Agent Prompt

The prompt supplies distilled Indicator, Pattern and Trend reports with variant-enabled factor and sentiment sections. It requests comparison of trend/key levels, candles, momentum and enabled quantitative/text evidence, followed by a forced LONG/SHORT forecast for the net next-open/target-close cycle. The output contract comprises direction, horizon, confidence, risk/reward, supporting/opposing evidence and justification. General instructions remain in Base, while both factor and sentiment sections are removed. Main compact factor text and factorial five-row text are different families. Section 4 and Figure 4 describe these distinctions; the exact archived request is authoritative.

Appendix A.2. Pattern Agent Prompt

The visual input is the archived candlestick PNG. The requested report covers pattern, confidence, directional bias, evidence, key candles and implication. The body-colour convention is green for Close greater than Open and red otherwise, including doji. This instruction and the deterministic consistency flag do not establish that the model grounded its answer in local chart evidence.

Appendix A.3. Trend Agent Prompt

The trend input is a separate PNG with support/resistance annotations. The requested report covers direction, support, resistance, line slope, price relative to support, detailed analysis, forecast and confidence. Support is blue and resistance red in the corrected input. Actual Vietnamese wording, inherited high-frequency language and unknown semantic compliance remain documented; an English summary does not remove those limitations.

Appendix A.4. Indicator Agent Prompt

The software contains an optional LLM reporting path, but the evaluated backtest returns a deterministic report without that LLM call. The five votes, counts and heuristic confidence are generated numerically. Treating the optional prompt as an observed study request would misdescribe both the input contract and call cost. The rules used here are in Appendix D.

Appendix A.5. Sentiment LLM Prompt (Subroutine of Alpha Agent)

The frozen sentiment path uses saved classifier scores and aggregate/count fields. It does not establish that an LLM read a peer/article corpus in this experiment. Scorer identity, the probability-difference score, decay and unavailable peer fields are specified in Section 3. Optional crawler/LLM capabilities in the application are not substituted for missing frozen inputs.

Appendix A.6. Alpha Agent LLM Reasoning Prompt

An optional live-mode reasoning prompt exists, but the frozen numerical/report path bypasses an additional Alpha reasoning call. The enabled factor rows and skill fields are delivered to Decision using the recorded family. We consequently describe an Alpha role rather than claiming a separate model interpreted every selected factor before Decision.

Appendix B. LangGraph State Schema

The declared `IndicatorAgentState` is a shared `TypedDict`; the backtest wrapper and individual nodes determine which fields are actually populated. Inputs include `kline_data`, `time_frame`, `stock_name`, `is_backtest`, `point_in_time_df`, `as_of_date`, language and the two image payloads. Report fields are `indicator_report`, `alpha_report`, `pattern_report`, `trend_report` and `sentiment_report`; sentiment data/normalizer fields and message history are distinct from those reports. Final output fields include `decision_prompt` and `final_trade_decision`.

Declared RSI/MACD/Stochastic/ROC/Williams, ATR, Bollinger and ADX/DI arrays do not imply that every array is forwarded by the deterministic backtest Indicator node or that each is a separate vote. Likewise, a sentiment schema comment does not establish that its peer or relative fields were observed. The forecasting target is the net Open-to-Close cycle in Eq. 4; it is separate from the state schema's field names. The evaluated pair shares upstream report parents, then uses variant-specific Decision composition and separate ledgers. The full schema listing and wrapper provenance are retained in source rather than reproduced as a misleading complete runtime trace.

Appendix C. Custom Alpha Formula Implementation

The custom formulas below specify native handlers and distinguish their preprocessing from prototype/fallback implementations. "Custom" describes their role in this implementation; it is not a validated novelty or profitability claim. The following formulas precede native global normalization and IC orientation. Handler identities are not the rank of a selected display row.

Table C.1: Custom native factor definitions and implementation distinctions.

Before global normalization and IC orientation

ID	Native handler expression	Distinction
FDM	$\tanh(0.5\, a_t V_t / ADV_{20,t})$	Native uses adjusted one-bar return, not the MACD display template.
SFA	$\tanh(1.5 \tanh(H_t/(0.005C_t + \epsilon)) + 0.5a_t)$	Native uses MACD histogram and return, not sentiment; prototype has a distinct sentiment branch.
LVR	$\tanh(-0.8\, z_{20}((C_t - SMA_{5,t})/C_t))$	Rolling mean/SD conventions and partial windows are implementation-specific.
BFE	$\tanh(1.5(\%B_t - 0.5)\, VolZ_{5,t})$	Negative VolZ reverses the sign; zero cancels. No squeeze or volume-expansion gate.
OFE	$\tanh(3a_t\, CP_t)$	OHLCV proxy, not measured signed order flow.

a_t is one-bar pct-change divided by rolling 20-bar sample SD (minimum five); H_t is MACD(12,26,9) histogram; CP_t is close location relative to the bar midpoint/range; ADV is mean Volume, not dollar volume. Guarded denominators and fills follow the registry. BFE registry sample-SD preprocessing differs from the rounded/population-SD prototype. Six selected metadata slots differ from their handler description, without the literal wrong formula in the affected prompts. A display rank of four does not identify BFE. Full expressions, guards and counterexamples are in the companion registry.

Appendix C.1. Flow-Driven Momentum (FDM)

FDM multiplies volatility-adjusted one-bar return by a volume/ADV20 ratio before the scaled tanh. The return adjustment uses rolling sample SD with its implementation-specific minimum observations. It is an OHLCV proxy, not directly measured order flow.

Appendix C.2. MACD/Return Proxy (SFA)

The native SFA handler combines a normalized MACD histogram and adjusted return. Its name/description and the earlier MACD-Flow display template do not substitute for the handler. No sentiment variable enters this native formula; the prototype has a separate sentiment branch.

Appendix C.3. Liquidity Void Reversion (LVR)

LVR applies a rolling z-score to the Close–SMA5 deviation divided by Close, with a negative multiplier before tanh. Guarded denominators, rolling windows, partial samples and sample-SD conventions are retained from the implementation.

Appendix C.4. Bollinger Squeeze & Flow (BFE)

BFE uses $\tanh(1.5(\%B - 0.5)VolZ_5)$. Negative volume z-score reverses the sign, and zero cancels it. There is no volume-expansion or squeeze gate in this formula. Registry preprocessing uses sample SD and differs from the rounded/population-SD prototype; they are not treated as interchangeable.

Appendix C.5. Order Flow Exhaustion (OFE)

OFE combines adjusted return with the close location relative to the bar midpoint/range. The name does not authenticate signed trade flow. Its guarded OHLCV expression and subsequent native normalization remain part of the actual signal definition.

Appendix D. Indicator Signal Classification Logic

Table D.1: Indicator signal rules and missing-data conventions.

Five deterministic votes

Signal	Priority and boundaries	Qualification
MACD	Previous strict/current inclusive crossing first; otherwise histogram sign and change.	12/26/9; no Stochastic histogram.
RSI	≥ 70: bear; ≤ 30: bull; $55 \leq x < 70$: bull; $30 < x \leq 45$: bear.	45–55 uses change against the third retained sample.
Stochastic	Crossing precedes ≥ 80 bear / ≤ 20 bull; interior fallback retained.	Legacy equality/zero-D truthiness is disclosed.
Williams %R	≥ -20: bear; ≤ -80: bull; otherwise retained-sample direction.	Not a volatility vote.
ROC	$> +0.5\%$: bull; $< -0.5\%$: bear; equality neutral.	Thresholds follow four-decimal rounding.

None/nonfinite values are removed, zero retained; rounding is to four decimals before classification. Retained samples need not be consecutive if missing. Missing/error gives a neutral vote within denominator five. Directional plurality, ties MIXED, and confidence are heuristics, not calibrated probabilities.

Appendix D.1. MACD Classification

Previous strict/current inclusive line crossing is checked before histogram sign/change. Parameters are 12/26/9. A Stochastic oscillator has no MACD histogram; the two rules must not be conflated.

Appendix D.2. RSI Classification

The ordered extreme branches (≥ 70 bearish, ≤ 30 bullish) precede interior trend bands. Thus $55 \leq RSI < 70$ is bullish and $30 < RSI \leq 45$ bearish; the middle interval uses change against the third retained observation. The ordered branches make the extreme and interior regions mutually exclusive.

Appendix D.3. Stochastic Oscillator Classification

Crossing has priority over overbought/oversold thresholds. Legacy equality and zero-D truthiness behaviour is retained and disclosed rather than replaced retrospectively with an idealized oscillator rule.

Appendix D.4. Williams %R Classification

Values ≥ -20 are bearish and ≤ -80 bullish before the interior retained-sample direction rule. This is a bounded momentum indicator, not a volatility measure.

Appendix D.5. ROC Classification

After four-decimal rounding, ROC greater than $+0.5\%$ is bullish, below -0.5% bearish, and equality is neutral.

Appendix D.6. Consensus Rule

Nonfinite samples are removed while zero is retained; retained observations need not be consecutive. Missing/error votes are neutral in denominator five. Directional plurality, a mixed tie and heuristic confidence are distinct from a requirement that every directional vote have a strict majority. None is a calibrated forecast probability.

Appendix E. Software Dependencies and Reproducibility

The research runtime is Python 3.13.5 with pinned numerical, graph, model-client and chart dependencies recorded in the reproducibility package. Indicator calculations use TA-Lib; chart images and numerical diagnostics have their own source hashes. Raw actual prompts, callback/error records, retry/checkpoint state, provider aliases and model/scorer assets are retained separately from scientific companion tables. Local saved-prediction replay reproduces metric calculations, not newly served model outputs or historical news availability.

The operational installation, Flask/JSON handling, tokenizer configuration and optional endpoint routing instructions remain in the repository. Their existence does not establish that each backend was used in the frozen study. Public distribution remains conditional on rights and a real persistent identifier. No credentials or private review document belong in a public reproducibility package.

Appendix F. Additional Account and Symbol Results

These tables use the same final supports and unchanged predictions as the main results. Supplement S1 gives every baseline by symbol and both classes; S2 contains all execution rows/contrasts; S3 the complete numeric/random family; S4 all temporal/regime/HAC diagnostics. They are scientific companion tables, distinct from raw execution logs and implementation files. No additional market observation is created by these alternative tabulations.

Table F.1: Full and Base symbol-level results on their common historical support.

Percent except MCC

Symbol/variant	UP/DOWN	Acc.	BA	MCC	R_U	R_D
BHN/Full	1/5	100.00	100.00	1.000	100.00	100.00
BHN/Base	1/5	66.67	80.00	0.447	100.00	60.00
CMG/Full	3/9	66.67	44.44	-0.174	0.00	88.89
CMG/Base	3/9	75.00	61.11	0.258	33.33	88.89
FPT/Full	2/10	66.67	40.00	-0.200	0.00	80.00
FPT/Base	2/10	50.00	30.00	-0.316	0.00	60.00
HVN/Full	2/10	66.67	60.00	0.158	50.00	70.00
HVN/Base	2/10	66.67	60.00	0.158	50.00	70.00
MBB/Full	5/7	50.00	48.57	-0.029	40.00	57.14
MBB/Base	5/7	58.33	55.71	0.120	40.00	71.43
MWG/Full	4/8	58.33	56.25	0.120	50.00	62.50
MWG/Base	4/8	50.00	50.00	0.000	50.00	50.00
VCB/Full	4/8	58.33	50.00	0.000	25.00	75.00
VCB/Base	4/8	50.00	37.50	-0.316	0.00	75.00
VJC/Full	4/8	58.33	43.75	-0.213	0.00	87.50
VJC/Base	4/8	50.00	37.50	-0.316	0.00	75.00
VNM/Full	0/12	83.33	NA	NA	NA	83.33
VNM/Base	0/12	83.33	NA	NA	NA	83.33

BHN contributes six eligible units, other symbols twelve; NA retains missing-class denominators. S1 additionally contains all nine classifiers for every symbol and both-class precision/recall/F1, with pooled and macro summaries.

Table F.2: All twelve account-execution sensitivities on unchanged predictions.

Means over 51 accounts per variant

Scenario	Base (%)	Full (%)	ΔFull (pp)	Stock	Pending
reference	-0.750	-0.797	0.000	0	0
pending_t2	-0.750	-0.797	0.000	0	0
advance_on_demand	-0.750	-0.797	0.000	0	0
advance_auto	-0.803	-0.840	-0.043	0	0
lower_cost	-0.526	-0.612	0.185	0	0
higher_cost	-1.249	-1.211	-0.414	0	0
sell_charge_10bp	-0.805	-0.843	-0.046	0	0
no_fill_proxy	-0.659	-0.803	-0.006	0	0
delayed_proxy	-0.621	-0.766	0.030	0	0
lot100_cap1pct	-0.602	-0.745	0.052	0	0
combined	-1.074	-1.166	-0.369	0	0
all_entries_missed	0.000	0.000	0.797	0	0

Stock/Pending count Full accounts with residual shares/receivables; zero Full residuals do not imply none for other strategies. All 6,120 account executions are in the companion results. Fee/slippage, advance fees, sell charge, lots and capacity are assumptions. No-fill/delay proxies do not measure queue fills; all entries missed is not a mathematical worst bound.

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